

Dakshi Goel

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EDUCATION

University of Maryland (College Park, MD)

Sept 2025 – May 2027

M.S. in Data Science | GPA: 4.0/4.0

Indian Institute of Technology (Jammu, India)

July 2020 – May 2024

B.Tech in Electrical Engineering and Computer Science | GPA: 3.48/4.0

- Awarded **Smt. Godavari Devi Award** for **highest** CGPA in Electrical Engineering [🔗](#)
- **2nd place** (Top 0.1% of 9,000 teams nationally) in Flipkart Grid Robotics Challenge [🔗](#)

WORK EXPERIENCE

Teaching Assistant | University of Maryland (College Park)

Feb 2026 - Current

- Assisting and grading 30+ students for the course Introduction to Data Science in the Computer Science department

AI Product Associate | The Real Brokerage (Miami, Florida)

Dec 2024 - Aug 2025

- Optimized a hybrid vector search and a reranking strategy, reducing query latency by 11%.
- Analyzed 10K+ real-time conversations using LangFuse to diagnose intent routing failures, collaborating with PMs.
- **Won** AI Hackathon (among 200+ employees) by building a data annotation tool to accelerate model training.

AI Developer | Mozrest (London, UK)

Dec 2023 - Dec 2024

- Architected an agentic chatbot using GPT-3.5 Turbo and LangGraph for restaurant booking, integrated Stripe payment gateway, and deployed it on AWS.
- Created a restaurant review response generation chatbot using few-shot prompting on GPT3.5, deployed by AWS Lambda.
- Built a restaurant sales forecasting model, testing with ARIMA and LSTM, achieving **RMSE 7.2 on 50+** daily bookings.

AI Innovation Intern | Infineon (Bangalore, India)

July 2023 - Dec 2023

- Developed RAG pipeline for QnA on semiconductor datasheets using LLaMA-13B+FAISS, achieving **0.65** BLEU score.
- Implemented T5 transformer and performed recursive chunking for translation and summarization on semiconductor data.

Machine Learning Intern | LTIMindtree (Bangalore, India)

May 2023 - July 2023

- Implemented LLaMA-2 13B for code generation on scraped Snowpark data, reducing hallucinations by **38%**.
- Built a Streamlit chatbot interface and integrated it with FastAPI, increased user engagement by 20%

PROJECTS

LLM Fingerprinting: Identifying the Author Through Text Analysis | UMD [🔗](#)

Sept 2025 – Dec 2025

- Achieved 82% classification accuracy across 7 LLMs and a human author using DeepNN for the MGTBench dataset.
- Extracted 10+ semantic and stylometric features (Langdetect, NER, NLTK), identifying perplexity as key feature.

Underwater 3D Reconstruction | Perception and Robotics Group | UMD

Sept 2025 -Current

- Parallelized training of SOTA 3D reconstruction transformers across 4 GPUs using Distributed Data Parallel.
- Modified model architectures to generate metric depth maps from underwater images using the depth-pro model.

Audio quality assessment | IIT Jammu [🔗](#)

Dec 2023 – Jan 2024

- Classified user-generated audio into score (1-5) over 2075 audio samples, adding a double encoder transformer architecture.
- Performed feature selection and achieved **PLCC 0.828, SRCC 0.823, KRCC 0.629**, surpassing baseline model.

Age and gender prediction from surveillance video | Inter-IIT TechMeet, KGP [🔗](#)

Dec 2021 – Jan 2022

- Developed an ESRGAN-based data pipeline for age/gender prediction from low-quality video feed.
- Integrated RetinaFace and regression-classification hybrid model for face detection, improving F1 by **0.1**.

PUBLICATIONS

Illumination aware multi-scale attention fusion model for underwater image enhancement [🔗](#)

IEEE: Artificial Intelligence | Indian Patent (No.202411053490)

- Designed encoder-decoder network with attention-based fusion, achieved PSNR of **31.83 dB** and SSIM of **0.948**.
- Integrated a multiscale, LAB/LCH color loss function, enhancing image color quality with **UIQM 4.284**.

Remote sensing data processing using CNNs for mapping alteration zones [🔗](#)

MIGARS : Conference on Machine Intelligence for GeoAnalytics and Remote Sensing

- Performed data preprocessing of on hyperspectral data using PCA to 3 critical identify mineral zones.
- Trained CNN to identify 3 alteration mineral zones, outperforming SVM and MLP, achieving **98.4% accuracy**.

SKILLS

Programming & Deployment: Python, C++, SQL, MLflow, Docker, FastAPI, AWS, Git, CI/CD Pipelines, Linux.

Relevant Coursework: Machine Learning, Deep Learning, Image Processing, Data Science, Linear Algebra, Algorithms

GenAI & NLP: LangChain, LangGraph, LangFuse, LangSmith PyTorch, TensorFlow, Transformers (Hugging Face), Llama-3, OpenAI, NLTK, FAISS

Data & Prototyping: PostgreSQL, Pandas, Numpy, Streamlit, Matplotlib, Seaborn